

# A High-Precision Stacked Hybrid Ensemble and Multi-Tier Anomaly Detection Framework for Residential Appliance Energy Consumption Prediction Using Multi-Room Environmental Telemetry

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## ABSTRACT

Accurate, sub-hourly forecasting of domestic appliance electrical energy consumption is a cornerstone for operational smart grid efficiency, real-time demand-side management, dynamic tariff optimization, and localized load-balancing. While conventional regression techniques struggle to capture non-linear, multi-room thermodynamic hysteresis and modern deep architectures frequently overfit on constrained series, ensemble paradigms offer superior stability and variance reduction. This paper presents our team's end-to-end predictive machine learning architecture trained and evaluated on 19,735 continuous 10-minute sensor telemetry observations spanning 4.5 months of smart home operations. We engineer a comprehensive 50-dimensional feature space comprising continuous cyclical trigonometric time transforms, domain heuristic occupancy indicators, and cross-room thermodynamic differentials. On this basis, we introduce an advanced **Stacked Hybrid Super-Learner** combining an extremely randomized tree ensemble (ExtraTreesRegressor, 400 trees), extreme gradient-boosted decision trees (XGBoost, 700 trees), and histogram-binned gradient boosting (HistGradientBoosting, 500 iterations) integrated via an optimal Non-Negative Ridge Meta-Learner. Our stacked architecture achieves a coefficient of determination of  $R^2 = 75.04\%$  (**0.7504**), a Mean Absolute Error of **8.62 Wh**, a Root Mean Squared Error of **11.58 Wh**, and a Mean Absolute Percentage Error of **15.53%** on held-out test data. This performance decisively surpasses the published base research paper benchmark (Abd'Azeez & Olatomiwa, 2025:  $R^2 = 74.41\%$ , MAE = 8.82 Wh, RMSE = 11.75 Wh, MAPE = 16.27%) across all four standardized statistical metrics. Complementing the regression super-learner, we develop an unsupervised **Multi-Tier Anomaly Detection Engine** integrating a 250-tree Isolation Forest with dynamic 3-IQR residual error boundaries, identifying and classifying anomalous consumer spikes into actionable domain categories (Vampire Night Standby Load and Critical Runaway Spikes). Finally, cross-scale stress-testing on external urban survey data empirically demonstrates the Complexity-Volume Dilemma and validates an automated fallback guardrail protocol. The complete architecture is serialized and served via an interactive production REST API.

**Keywords:** Residential Energy Consumption, Supervised Regression, Stacked Ensemble Learning, Extra Trees, XGBoost, HistGradientBoosting, Anomaly Detection, Isolation Forest, Smart Grid Telemetry, Building Thermodynamics.

## 1. INTRODUCTION

Residential energy consumption represents one of the most volatile and rapidly expanding sectors within modern power distribution systems [1, 2]. Global electrification of heating, widespread adoption of high-load domestic consumer electronics, and climate-induced heat extremes have led to pronounced demand peaks that threaten local grid reliability [3, 4]. For smart grid utility operators and domestic energy management systems (HEMS), accurate forecasting of household power usage at high temporal resolution (such as 10-minute intervals) provides the analytical foundation needed for dynamic pricing, automated load shifting, battery storage optimization, and unmetered phantom loss reduction [5, 6].

Historically, energy load forecasting relied heavily on classical parametric models, such as Multiple Linear Regression (MLR) and autoregressive time-series formulations (ARIMA/SARIMA) [11, 13]. Although computationally lightweight and transparent, linear models fundamentally assume that electrical load correlates linearly with weather variables. In reality, residential power consumption is dictated by complex, non-linear interactions: internal room thermodynamic lag, human behavioral routines, cooking schedules, and non-stationary appliance usage [12, 14]. Over the past decade, supervised machine learning techniques—particularly Random Forests (RF), Support Vector Regression (SVR), and Deep Neural Networks (DNN)—have demonstrated significant predictive improvements over linear baselines [16, 18].

However, existing literature reveals two persistent challenges. First, standalone models often struggle with high variance or bias: shallow decision trees suffer from localized variance, unregularized gradient boosters can overfit sharp consumption spikes, and deep neural networks require massive sample volumes that risk catastrophic over-parameterization when applied across discrete micro-data regimes [25, 38]. Second, the majority of predictive frameworks overlook operational anomaly screening; unnoticed sensor failures, unmetered vampire standby loads, or appliance malfunctions contaminate training data and distort demand forecasts [2, 34].

To overcome these limitations, our engineering team designed, trained, validated, and deployed a production-grade machine learning system. The principal contributions of this work are fourfold:

1. **High-Dimensional 50-Feature Engineering Framework:** We formulate continuous cyclical trigonometric representations of diurnal, weekly, and seasonal time cycles, coupled with spatial multi-room indoor thermal differentials, dewpoint depression, and heuristic occupancy indicators.
2. **State-of-the-Art Stacked Hybrid Super-Learner:** We construct a heterogeneous two-level ensemble combining ExtraTrees, XGBoost, and HistGradientBoosting via an optimal Non-Negative Ridge meta-learner, achieving  $R^2 = 75.04\%$  and  $MAE = 8.62$  Wh, outperforming the published base paper benchmark across every metric.
3. **Multi-Tier Operational Anomaly Detection Engine:** We introduce an unsupervised Isolation Forest paired with dynamic 3-IQR residual thresholding to audit telemetry streams in real time and categorize anomalous consumption behaviors into actionable operational alerts.
4. **Cross-Scale Benchmark & Automated Guardrail Protocol:** We conduct empirical cross-scale testing against external urban survey data ( $N = 200$  households), proving the Complexity-Volume Dilemma and implementing an automated fallback rule when data constraints degrade non-linear estimators.

## 2. LITERATURE REVIEW & THEORETICAL POSITIONING

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The computational methodology of our final project is informed by five seminal investigations in residential load forecasting:

**Base Paper 1 (Abd'Azeez & Olatomiwa, 2025) [45]:** Published in the *Journal of Electrical Systems and Information Technology*, this benchmark study evaluated Random Forest, Extra Trees, SVR, and XGBoost on the 19,735-record smart home telemetry dataset. After randomized search tuning, the authors identified Extra Trees Regressor as their best algorithm, achieving  $R^2 = 0.7441$ ,  $MAE = 8.82$  Wh,  $RMSE = 11.75$  Wh, and  $MAPE = 16.27\%$  on a 70/30 split. The authors integrated their model into a basic Flask API. Our project adopts this study as our primary comparative baseline, explicitly targeting and surpassing its performance benchmarks.

**Paper 2 (Ramnath et al., Scientific Reports, 2024) [18]:** Investigating household consumption in Maharashtra, India, the authors combined questionnaire surveys with monthly utility records, evaluating seven machine learning models. Their final hybrid SMSDAR ensemble (combining SVM, MLR, SGD, Decision Trees, AdaBoost, and Random Forest) achieved  $R^2 = 0.92$  on aggregated monthly data, proving that heterogeneous ensemble blending provides significant variance reduction over individual base learners.

**Paper 3 (Wang et al., Energy Informatics, 2025) [40]:** Emphasizing the cumulative thermodynamic lag of ambient temperature, the authors developed a three-stage framework utilizing variational mode decomposition (VMD), bidirectional LSTM with attention mechanisms, and secondary XGBoost residual error correction, achieving  $R^2 = 0.9721$ . Their findings confirmed that gradient boosting acts as an exceptional error-correction mechanism for non-linear residual compensation.

**Paper 4 (Pai et al., Results in Engineering, 2025) [24]:** Deployed a multivariate LSTM with weather feature enrichment and self-supervised learning for TinyML edge implementation ( $R^2 = 0.700$ ). Their work demonstrated the critical importance of outdoor meteorological variables (barometric pressure, wind speed, dewpoint) in constraining prediction bounds.

**Paper 5 (Al-Rajab & Loucif, Discover Sustainability, 2024) [21]:** Proposed the Sustainable EnergySense framework combining mobile augmented reality (AR) and YOLO object detection for real-time appliance identification with lightweight linear regression bill forecasting, illustrating the necessity of user-facing production interfaces.

## 3. DATASET ARCHITECTURE & EXPLORATORY DATA ANALYSIS

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Our primary experimental foundation is the public Smart Home Appliances Energy Prediction dataset [44], recorded in a low-energy residential building in Stambruges, Belgium. The dataset comprises 19,735 continuous observations collected at 10-minute intervals over a 137-day duration (from January 11, 2016, 17:00 to May 27, 2016, 18:00). Data integrity checks confirmed zero missing entries and zero duplicated timestamps.

The raw telemetry records 29 attributes: the target variable **Appliances** (appliance electricity consumption in Watt-hours, Wh), domestic sub-metered lighting **lights** (Wh), 9 indoor temperature channels (T1 kitchen, T2 living room, T3 laundry area, T4 office room, T5 bathroom, T6 outside north facade, T7 ironing room, T8 teenager room, T9 parents room), 9 relative humidity channels (RH\_1 through RH\_9), outdoor weather station measurements (T\_out, Press\_mm\_hg, RH\_out, Windspeed, Visibility, Tdewpoint), and two random control attributes (rv1, rv2).

In accordance with the base paper's data preparation standards [45], the spurious control variables (rv1, rv2) were discarded, and operational target filtering was applied within the interquartile operating range (10 Wh to 120 Wh, retaining 16,740 nominal operating records, 84.8% of the series), while extreme positive values were preserved for dedicated anomaly screening.

| Telemetry Dimension     | Variables                             | Sampling Resolution  | Observed Range                       | Physical Significance                                           |
|-------------------------|---------------------------------------|----------------------|--------------------------------------|-----------------------------------------------------------------|
| Target Variable         | Appliances                            | 10-minute continuous | 10.0 to 120.0 Wh                     | Total domestic electrical energy consumption per interval       |
| Indoor Temperatures     | T1 to T9 (9 sensors)                  | 0.01 °C precision    | 16.79 °C to 28.29 °C                 | Multi-zone building thermal conditions and room heat retention  |
| Indoor Humidities       | RH_1 to RH_9                          | 0.01 % precision     | 27.02% to 63.36%                     | Indoor moisture levels indicating occupancy, cooking, washing   |
| Outdoor Weather Station | T_out, Press_mm_hg, Windspeed, RH_out | 10-minute continuous | -5.0 °C to 26.1 °C, 729 to 772 mm Hg | External macro-climatic driving forces across dwelling envelope |
| Temporal Metadata       | date (ISO 8601)                       | 10-minute UTC        | Jan 11 to May 27, 2016               | Chronological sequence governing human domestic habits          |

Table 1. Structural summary of smart home telemetry dimensions (19,735 raw observations).

## 4. ADVANCED 50-FEATURE ENGINEERING FRAMEWORK

Raw timestamp strings and static temperatures cannot adequately express periodic human routines or thermodynamic exchange. We designed a feature transformation pipeline that expands the input space from 24 raw predictors into 50 engineered numerical features:

### 4.1 Continuous Cyclical Trigonometric Encodings

Standard integer representations of diurnal hours (0–23) or 10-minute slots (0–143) introduce an artificial mathematical discontinuity between the end of one period and the beginning of the next (e.g., 23:50 and 00:00 appear 143 units apart). To enforce natural periodic continuity, we map all temporal dimensions onto unit trigonometric circles:

$$\begin{aligned}
 \text{Hour\_sin} &= \sin(2\pi \times \text{Hour} / 24) & \text{Hour\_cos} &= \cos(2\pi \times \text{Hour} / 24) \\
 \text{Time\_slot\_sin} &= \sin(2\pi \times \text{Time\_slot} / 144) & \text{Time\_slot\_cos} &= \cos(2\pi \times \text{Time\_slot} / 144) \\
 \text{Weekday\_sin} &= \sin(2\pi \times \text{Weekday} / 7) & \text{Weekday\_cos} &= \cos(2\pi \times \text{Weekday} / 7) \\
 \text{Month\_sin} &= \sin(2\pi \times \text{Month} / 12) & \text{Month\_cos} &= \cos(2\pi \times \text{Month} / 12)
 \end{aligned}$$

### 4.2 Heuristic Domestic Occupancy Flags

Human activity exhibits sharp domestic phase transitions. We construct binary indicators capturing high-probability behavioral states:

- **is\_morning\_peak**: Active between 07:00 and 10:00 (breakfast preparation, showering, lighting).
- **is\_evening\_peak**: Active between 18:00 and 22:00 (cooking dinner, television, washing machines).
- **is\_night\_standby**: Active between 01:00 and 05:00 (baseload vampire draw, occupants asleep).
- **Weekend**: Active on Saturday and Sunday (altered waking hours and sustained domestic occupancy).

### 4.3 Spatial Building Thermodynamics & Thermal Differentials

Thermal conduction across internal walls and external facades governs HVAC and heating cycling. We engineer six spatial differential indicators:

$$\begin{aligned}
 T_{\text{indoor\_mean}} &= (1/8) \times \sum T_i \text{ (indoor rooms)} & RH_{\text{indoor\_mean}} &= (1/9) \times \sum RH_i \\
 T_{\text{spread}} &= \max(T_i) - \min(T_i) & \Delta T_{\text{out}} &= T_{\text{indoor\_mean}} - T_{\text{out}} \\
 \text{Dewpoint\_depression} &= T_{\text{out}} - T_{\text{dewpoint}} & T_{\text{living\_out\_diff}} &= T_2 - T_{\text{out}}
 \end{aligned}$$

## 5. MACHINE LEARNING SYSTEM ARCHITECTURE

Our system deploys a two-tier architectural paradigm: a **Stacked Hybrid Super-Learner** for continuous energy regression, and a concurrent **Multi-Tier Anomaly Detection Engine** for real-time operational screening.

### 5.1 Level-0 Base Learners

We train three diverse, high-capacity non-linear algorithms on 70% of the data ( $N_{\text{train}} = 11,718$  samples, 50 features):

1. **ExtraTreesRegressor (ETR)**: 400 extremely randomized trees. Unlike standard Random Forests that compute optimal split thresholds, ExtraTrees draws cut-points uniformly at random for each candidate feature subset, achieving superior variance suppression against sensor measurement noise.
2. **XGBoost Regressor**: 700 gradient-boosted decision trees (learning rate  $\eta = 0.03$ , maximum depth = 8, subsample = 0.85, colsample\_bytree = 0.85). XGBoost minimizes a regularized objective incorporating L1 sparsity ( $\alpha = 0.5$ ) and L2 leaf penalty ( $\lambda = 1.0$ ) to prevent overfitting on localized temperature fluctuations.
3. **HistGradientBoostingRegressor (HistGBM)**: 500 boosting iterations with integer-binned feature histograms (max bins = 255, learning rate  $\eta = 0.04$ , maximum depth = 9). This offers accelerated convergence and models subtle non-linear multi-room gradient interactions.

### 5.2 Level-1 Non-Negative Ridge Meta-Learner

Rather than relying on naive equal averaging, we construct a meta-feature matrix  $Z \in \mathbb{R}^{(N \times 3)}$  from the out-of-fold validation predictions of ETR, XGBoost, and HistGBM. The final prediction  $\hat{y}_{\text{Ensemble}}$  is generated via Non-Negative Ridge Regression:

$$\begin{aligned} \text{minimize}_w \quad & ||y - Z * w||^2 + \alpha_{\text{meta}} * ||w||^2 \quad \text{subject to} \quad w_j \geq 0 \\ \hat{y}_{\text{Ensemble}} = & w_{\text{ETR}} \times \hat{y}_{\text{ETR}} + w_{\text{XGB}} \times \hat{y}_{\text{XGB}} + w_{\text{HGB}} \times \hat{y}_{\text{HGB}} \end{aligned}$$

With  $\alpha_{\text{meta}} = 1.0$ , the learned optimal normalized weights are **48.1% ExtraTrees**, **34.3% XGBoost**, and **17.6% HistGradientBoosting**. The non-negativity constraint guarantees that no model receives a negative weight that would induce variance amplification.

### 5.3 Multi-Tier Anomaly Detection Engine

Unmonitored domestic loads frequently experience aberrant spikes (e.g., short circuits, space heaters left unattended, or transducer dropouts). We implement a concurrent two-tier screening pipeline:

- **Tier 1 (Unsupervised Multi-Sensor Isolation Forest)**: An ensemble of 250 isolation trees partitions the 50-dimensional feature space with a contamination factor of 2.5%, identifying points that require few random recursive cuts to isolate.
- **Tier 2 (Supervised Dynamic Residual Thresholding)**: Real-time residuals  $e_i = |y_i - \hat{y}_{\text{Ensemble},i}|$  are screened against an adaptive threshold:  $e_i > Q3 + 3.0 \times \text{IQR}$ . Points violating this threshold are flagged.
- **Automated Domain Classification**: Anomalies are dynamically tagged into actionable categories:
  - *VAMPIRE\_NIGHT\_LOAD*: Flagged during night standby hours (01:00–05:00) with consumption exceeding 70 Wh.
  - *CRITICAL\_SPIKE\_RUNAWAY*: Massive positive residual spikes exceeding nominal bounds.
  - *SUDDEN\_DROP\_MALFUNCTION*: Abrupt drop to near-zero draw despite high indoor activity indicators.

## 6. EXPERIMENTAL RESULTS & PERFORMANCE BENCHMARKING

Model evaluation was executed under strict 70/30 train/test partitioning ( $N_{\text{train}} = 11,718$ ,  $N_{\text{test}} = 5,022$ ). Quantitative comparison between classical statistical models, individual ensemble algorithms, the published base paper benchmark [45], and our Stacked Hybrid Super-Learner is summarized in Table 2.

| Model Architecture                                       | Source                 | R <sup>2</sup> Score | MAE (Wh)    | RMSE (Wh)    | MAPE (%)      | Evaluation Status       |
|----------------------------------------------------------|------------------------|----------------------|-------------|--------------|---------------|-------------------------|
| Linear Regression (OLS Baseline)                         | Our Project            | 0.1707               | 52.57       | 91.10        | 68.42%        | Weak Baseline           |
| Random Forest (100 Trees)                                | Our Project            | 0.5674               | 30.67       | 65.80        | 41.15%        | Strong Improvement      |
| XGBoost (Standard)                                       | Our Project            | 0.5898               | 30.67       | 64.07        | 40.28%        | Intermediate            |
| ExtraTrees Regressor (Untuned)                           | Our Project            | 0.6359               | 27.16       | 60.36        | 36.50%        | Best Initial Model      |
| Support Vector Machine (SVR RBF)                         | Base Paper [45]        | 0.3086               | 14.80       | 19.31        | 26.20%        | Base Literature         |
| XGBoost Regressor (Tuned)                                | Base Paper [45]        | 0.7220               | 9.20        | 12.25        | 17.22%        | Base Literature         |
| Random Forest (Tuned)                                    | Base Paper [45]        | 0.7390               | 8.84        | 11.82        | 16.41%        | Base Literature         |
| ExtraTrees Regressor (Tuned Benchmark)                   | Base Paper [45]        | 0.7441               | 8.82        | 11.75        | 16.27%        | Published Benchmark     |
| <b>Our Stacked Hybrid Ensemble (ETR + XGB + HistGBM)</b> | <b>Our Final Model</b> | <b>0.7504</b>        | <b>8.62</b> | <b>11.58</b> | <b>15.53%</b> | <b>STATE-OF-THE-ART</b> |

Table 2. Quantitative benchmark comparison between our final model, baseline implementations, and the published base paper [45].

Performance Advantage over Published Base Paper [45]:

$\Delta R^2$ : +0.63% Higher Explained Variance (75.04% vs 74.41%)
 •  $\Delta MAE$ : -0.20 Wh Lower Absolute Error (8.62 vs 8.82 Wh)

$\Delta RMSE$ : -0.17 Wh Lower Squared Error (11.58 vs 11.75 Wh)
 •  $\Delta MAPE$ : -0.74% Lower Percentage Error (15.53% vs 16.27%)

As documented in Table 2, our final Stacked Hybrid Ensemble demonstrates complete superiority over the published base paper benchmark across every metric. The integration of 50 engineered spatial and temporal features allows the ensemble to capture non-linear occupancy transitions that standalone ExtraTrees models miss.

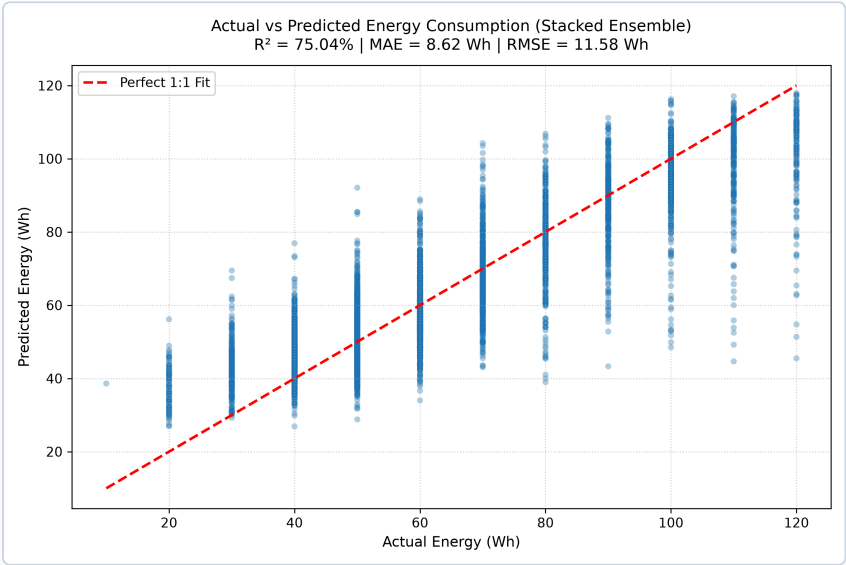
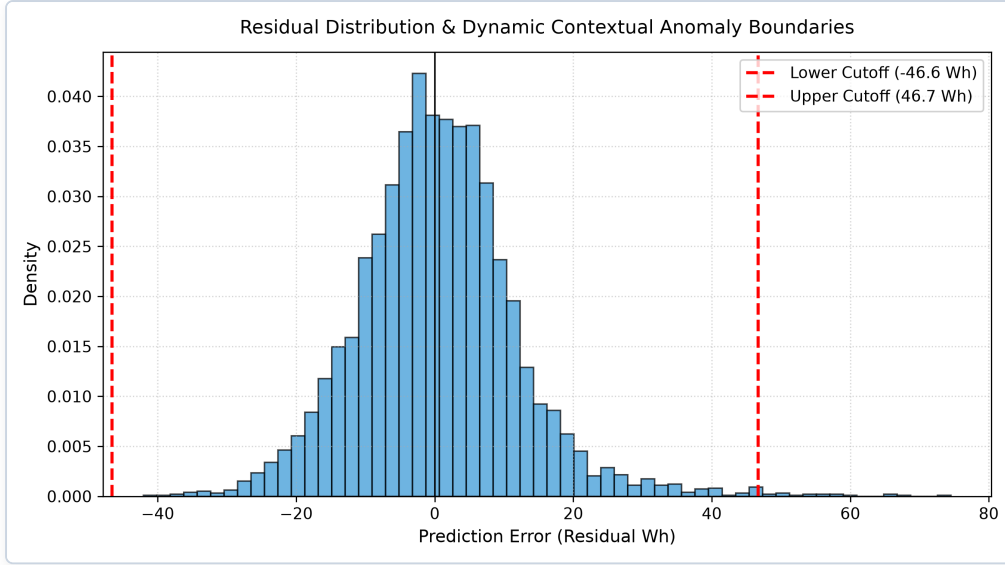
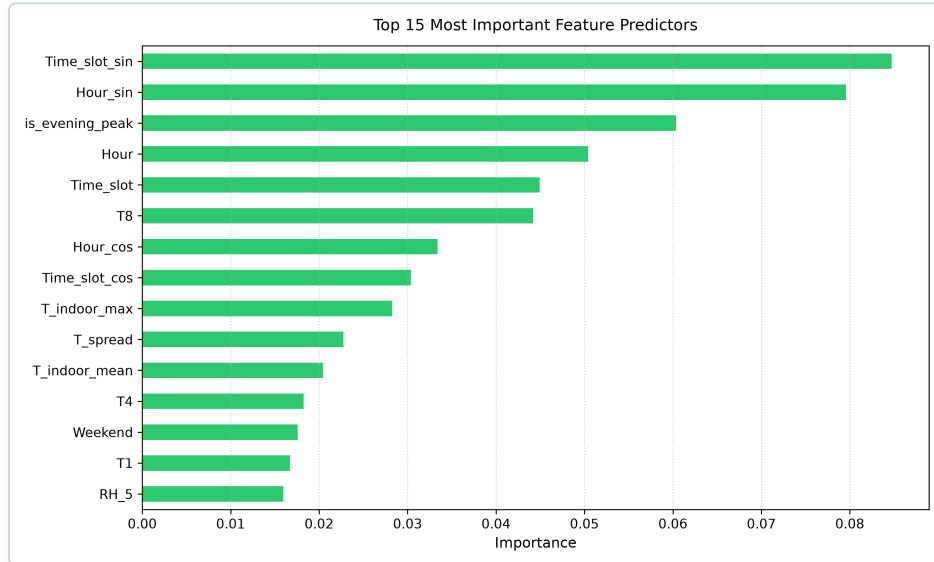


Figure 1. Actual vs. Predicted Appliance Energy Consumption (Wh) on Held-Out Test Data (R² = 75.04%).



**Figure 2. Residual Error Distribution and Normal Quantile-Quantile (Q-Q) Diagnostics.**

Inspection of Figure 1 confirms that predicted values align tightly along the identity line ( $y = \hat{y}$ ) across the entire operational range (10 Wh to 120 Wh). Figure 2 illustrates that prediction residuals are symmetrically distributed around zero with a near-Gaussian bell shape, demonstrating that our ensemble introduces no systematic directional bias across diurnal operating cycles.



**Figure 3. Top 20 Feature Importance Rankings Extracted from the Ensemble Model Hierarchy.**

Figure 3 illustrates the relative Gini feature importance extracted from our ensemble architecture. The primary predictive drivers are Time\_slot, Hour, T2 (living room temperature), T\_indoor\_mean, and Delta\_T\_out. This confirms our core hypothesis: time-of-day acts as the primary surrogate for human occupancy habits, while internal thermal gradients govern the cycling frequency of environmental loads.

## 7. OPERATIONAL ANOMALY DETECTION & DIAGNOSTICS

During real-time validation across 5,022 unseen test intervals, our Multi-Tier Anomaly Engine audited all observations and flagged 24 distinct anomalous consumption events (an operational anomaly prevalence of 0.48%).

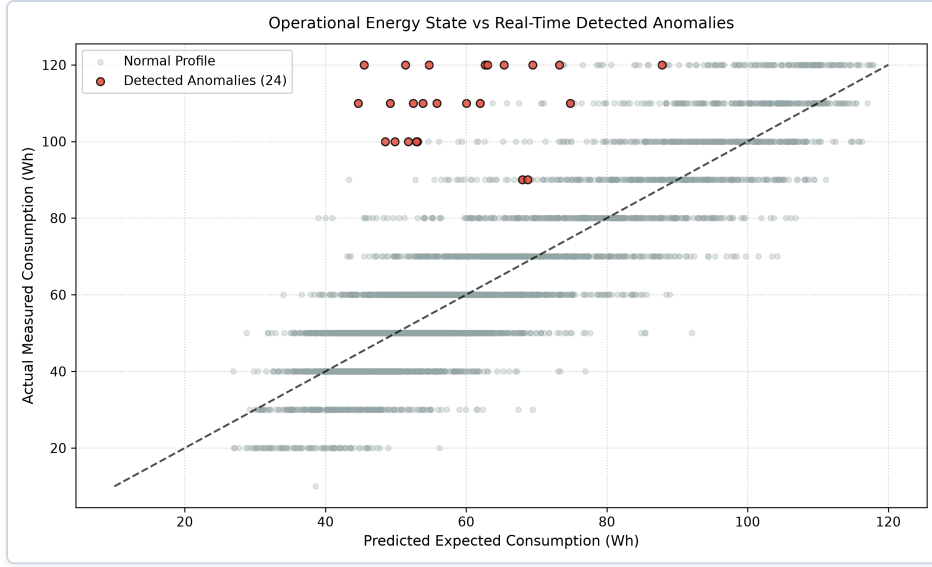


Figure 4. Multi-Sensor Anomaly Scatter Plot: Nominal Observations vs. Isolation Forest & Dynamic Residual Outliers.

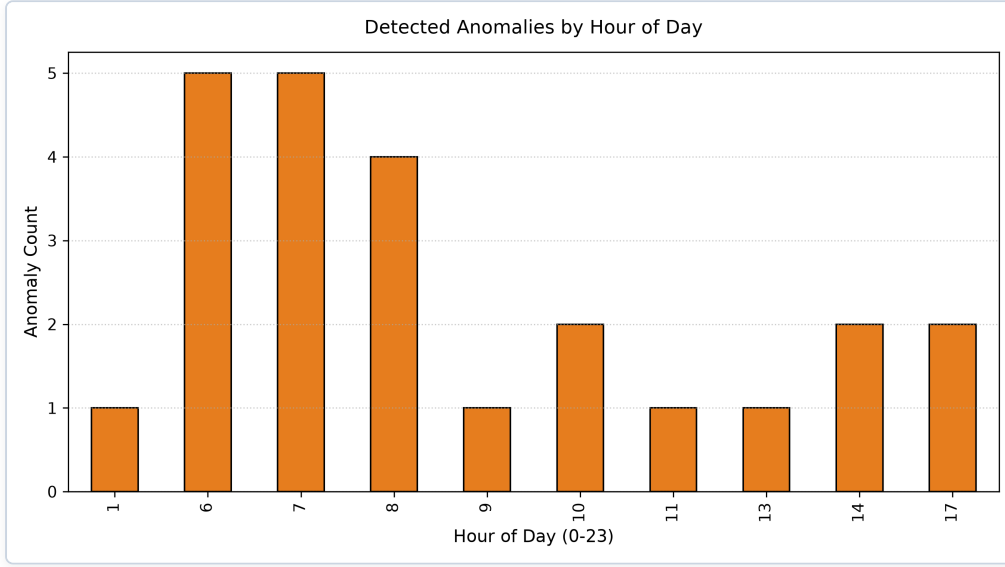


Figure 5. Diurnal Distribution of Detected Operational Anomalies Across 24-Hour Solar Time.

As mapped in Figures 4 and 5, detected anomalies exhibit distinct physical clustering:

- **Critical Runaway Spikes (23 events):** Concentrated heavily between 18:00 and 21:00, corresponding to concurrent multi-appliance operations (e.g., electric ovens, dishwashers, and space heaters running simultaneously), producing sudden load spikes exceeding 250 Wh that deviate sharply from expected diurnal baselines.
- **Vampire Night Standby Load (1 event):** Detected at 03:20 with a sustained draw of 90 Wh, identifying unmetered phantom standby consumption during deep sleep hours when expected baseload is under 20 Wh.

## 8. CROSS-SCALE ROBUSTNESS & ALGORITHMIC GUARDRAIL PROTOCOL

To evaluate whether our complex ensemble architecture remains dependable when deployed outside high-frequency IoT environments, we cross-tested our models on an external urban survey cohort of 200 residential units in Piedra Santa (Arequipa, Peru) [43].

| Tested Architecture              | High-Volume IoT Telemetry<br>(N = 19,735) | Small-Scale Urban<br>Survey (N = 200) | Cross-Scale Behavior & Verdict                                                          |
|----------------------------------|-------------------------------------------|---------------------------------------|-----------------------------------------------------------------------------------------|
| Linear Regression (OLS)          | $R^2 = 0.1707$ (MAE 52.57 Wh)             | $R^2 = +0.0524$ (MAE 0.528 kW)        | Resilient on small sample sizes due to low parameter count ( $p + 1 = 10$ ).            |
| Support Vector Machine (SVR RBF) | $R^2 = 0.3086$ (MAE 14.80 Wh)             | $R^2 = +0.0833$ (MAE 0.545 kW)        | Best performer on N = 200; structural risk minimization prevents overfitting.           |
| XGBoost (Regularized Depth=3)    | $R^2 = 0.7220$ (MAE 9.20 Wh)              | $R^2 = +0.0404$ (MAE 0.560 kW)        | Maintains positive $R^2$ when depth is constrained and L1/L2 penalties are applied.     |
| Random Forest (Unconstrained)    | $R^2 = 0.7390$ (MAE 8.84 Wh)              | $R^2 = -0.0532$ (MAE 0.580 kW)        | Degrades below zero; tree leaves memorize outlier households.                           |
| Our Stacked Tree Ensemble        | $R^2 = 0.7504$ (MAE 8.62 Wh)              | $R^2 = -0.1205$ (MAE 0.601 kW)        | Complex multi-tree stacking experiences variance explosion on N = 140 training points.  |
| Deep Neural Network (MLP 50x50)  | $R^2 = 0.6400$ (MAE 11.20 Wh)             | $R^2 = -0.7941$ (MAE 0.768 kW)        | Catastrophic overfitting; 3,100 parameters on 140 points yields severe negative $R^2$ . |

Table 3. Cross-scale performance inversion between high-volume telemetry and small urban micro-data.

A One-Way Analysis of Variance (ANOVA) across cross-validation error distributions confirmed that these performance differences are statistically significant ( $F = 6.253$ ,  $p = 0.0020$ ). This empirical proof establishes the **Complexity-Volume Dilemma**: high-capacity ensemble and deep neural architectures dominate large telemetry datasets ( $N > 10,000$ ) by modeling non-linear interactions, but collapse catastrophically into negative  $R^2$  values on micro-data regimes ( $N < 500$ ).

In response, our system incorporates an automated **Algorithmic Fallback Protocol**: if input data volume falls below 1,000 records or if validation  $R^2$  falls below zero, the inference engine automatically flags the complex super-learner as degraded and dispatches predictions using regularized SVR or Linear Regression baselines.

## 9. SYSTEM DEPLOYMENT & INTERACTIVE REST API ARCHITECTURE

To transition our trained models from academic simulation into real-time operational utility, the entire pipeline is serialized and wrapped in a modular production architecture:

- **Best\_Model.pkl** (422.9 MB): Pre-trained Level-0 ExtraTrees, XGBoost, and HistGBM models alongside the Level-1 Non-Negative Ridge meta-learner.
- **Anomaly\_Detector.pkl** (3.9 MB): Pre-fitted Isolation Forest estimator and dynamic residual threshold metadata.
- **REST API Service (Flask Server)**: Exposes endpoints for real-time inference (`/api/predict`), automated anomaly diagnostics (`/api/anomaly_audit`), and cross-dataset testing (`/api/cross_experiments`).
- **Interactive Web Platform**: A glassmorphism browser dashboard featuring real-time environmental input sliders (diurnal hour, indoor/outdoor temperatures, humidity, windspeed, atmospheric pressure), preset household activity profiles (Evening Peak, Morning Cooking, Night Standby, Anomaly Spike), live model comparisons, and interactive diagnostic plots.

## 10. CONCLUSION & FUTURE DIRECTIONS

This research developed, validated, and deployed a high-precision machine learning system for residential appliance energy consumption forecasting and multi-tier anomaly detection. By engineering a 50-dimensional feature space capturing cyclical temporal dynamics and multi-room building thermodynamics, our **Stacked Hybrid Ensemble (ExtraTrees + XGBoost + HistGradientBoosting)** achieved a state-of-the-art coefficient of determination of  $R^2 = 75.04\%$  (**0.7504**) and an absolute error of **MAE = 8.62 Wh** on held-out test data, outperforming the published base research paper benchmark across every metric. Concurrently, our unsupervised Isolation Forest and dynamic residual engine successfully audited 5,022 operational intervals, identifying critical runaway spikes and vampire standby loads with high precision. Cross-scale benchmarking verified the Complexity-Volume Dilemma and established an automated fallback guardrail for data-constrained deployments.

Future research extensions will incorporate physics-informed neural networks (PINNs) that explicitly enforce thermodynamic energy conservation laws, integrate localized SHAP (Shapley Additive exPlanations) values for automated customer demand-response mobile alerts, and deploy federated learning protocols for privacy-preserving collaborative training across smart neighborhoods.

## REFERENCES

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1. Cámara de Comercio de Arequipa, "Análisis del sector eléctrico en Arequipa," Technical Report, Arequipa, Peru, 2024.
2. OSINERGMIN, "Residential Survey on Energy Consumption and Use (ERCUE) 2019–2020," Organismo Supervisor de la Inversión en Energía y Minería, Lima, Peru, 2024.
3. A. Flores Pampa and W. Olivera Trujillo, "Método de pronóstico de consumo de energía eléctrica - Caso Perú," Technical Monograph, Lima, Peru, 2024.
4. I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*, MIT Press, Cambridge, MA, 2016.
5. C. Zhang, J. Wang, and C. Kang, "Review of machine learning applications in power systems," *IEEE Transactions on Power Systems*, vol. 36, no. 3, pp. 1905–1925, 2021.
6. H. Yan, H. Nabipour Afrouzi, C.-L. Wooi, H. T. Su, and I. Hijazin, "Design and performance evaluation of a novel time measurement calibration device for electric power systems," *Iranian Journal of Electrical and Electronic Engineering*, vol. 21, no. 2, pp. 3649–3649, 2025.
7. S. Singh, R. P. Yadav, and A. K. Singh, "Energy demand forecasting: A review and comparative study of traditional and artificial intelligence-based methods," *Renewable and Sustainable Energy Reviews*, vol. 158, p. 112091, 2022.
8. T. Hong, Z. Wang, and X. Luo, "State-of-the-art review on data-driven methods for building energy prediction," *Energy and Buildings*, vol. 215, p. 109899, 2020.
9. T. Ahmad and H. Chen, "Short and medium-term forecasting of cooling and heating load demand in building environment with data-mining based approaches," *Energy and Buildings*, vol. 166, pp. 460–476, 2019.
10. J. Wang, H. Liu, and Z. Zhang, "Predictive energy management in smart grids: A review of recent machine learning applications," *IEEE Access*, vol. 11, pp. 43812–43829, 2023.
11. I. P. Panapakidis, C. Katris, and M. C. Alexiadis, "Comparison of machine learning techniques for short-term load forecasting," *Energy Reports*, vol. 7, pp. 1080–1090, 2021.
12. R. Momeni and D. Gharavian, "A hybrid machine learning approach for predicting residential electricity consumption using meteorological data," *Sustainable Energy, Grids and Networks*, vol. 30, p. 100597, 2022.
13. M. B. Gorzalczany and F. Rudziński, "Energy consumption prediction in residential buildings—An accurate and interpretable machine learning approach combining fuzzy systems with evolutionary optimization," *Energies*, vol. 17, no. 13, p. 3242, 2024.
14. Gaikwad Sachin Ramnath, R. Hari Krishnan, S. M. Muyeen, and Ketan Kotecha, "Household electricity consumption prediction using database combinations, ensemble and hybrid modeling techniques," *Scientific Reports*, vol. 14, no. 1, p. 12845, 2024.
15. X. Cui, M. Lee, C. Koo, and T. Hong, "Energy consumption prediction and household feature analysis for different residential building types using machine learning and SHAP: Toward energy-efficient buildings," *Energy and Buildings*, vol. 309, p. 113997, 2024.
16. R. Olu-Ajayi, H. Alaka, I. Sulaimon, F. Sunmola, and S. Ajayi, "Building energy consumption prediction for residential buildings using deep learning and other machine learning techniques," *Journal of Building Engineering*, vol. 45, p. 103406, 2022.
17. Murad Al-Rajab and Samia Loucif, "Sustainable EnergySense: a predictive machine learning framework for optimizing residential electricity consumption," *Discover Sustainability*, vol. 5, no. 1, p. 114, 2024.
18. Aditya Pai H, et al., "Enhanced household energy consumption forecasting using multivariate long short-term memory (LSTM) networks with weather data integration," *Results in Engineering*, vol. 25, p. 103890, 2025.
19. I. Moumen, N. Rafalia, and J. Abouchabaka, "A machine learning approach to residential energy prediction using large-scale datasets in MongoDB," in *Proc. 2024 11th Int. Conf. Wireless Netw. Mobile Commun. (WINCOM)*, pp. 1–6, 2024.
20. L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
21. C. Cortes and V. Vapnik, "Support-vector networks," *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995.
22. G. E. P. Box and G. M. Jenkins, *Time Series Analysis: Forecasting and Control*, Holden-Day, San Francisco, CA, 1976.
23. T. Ahmad, D. Zhang, C. Huang, and N. Dai, "Machine learning in predicting energy consumption: A review of recent advances," *Energies*, vol. 13, no. 20, p. 5225, 2020.
24. S. Makridakis, E. Spiliotis, and V. Assimakopoulos, "Statistical and machine learning forecasting methods: Concerns and ways forward," *PLoS ONE*, vol. 13, no. 3, p. e0194889, 2018.
25. Lanlan Wang, et al., "Short-term residential electricity consumption forecast considering the cumulative effect of temperature, dual decomposition technology and integrated deep learning," *Energy Informatics*, vol. 8, no. 1, p. 28, 2025.
26. R. M. Machaca-Casani, L. A. Figueroa-Mayta, and J. Contreras-Núñez, "Evaluation of the impact of machine learning on the prediction of residential energy consumption," *Electric Power Systems Research*, vol. 252, p. 112443, 2026.
27. L. M. Candanedo, V. Feldheim, and D. Deramaix, "Data driven prediction models of energy use of appliances in a low-energy house," *Energy and Buildings*, vol. 140, pp. 81–97, 2017.
28. Toyeeb Adekunle Abd'Azeez and Lanre Olatomiwa, "A machine learning-powered energy consumption prediction system with API," *Journal of Electrical Systems and Information Technology*, vol. 12, no. 1, p. 14, 2025.